

scVI Human Validation — Consensus & Coordinator

Purpose

Coordinate the blinded validation, preserve both locked human records, and derive a consensus human reference standard. The machine-only crosswalk is intentionally kept outside Google Docs.

Reviewer workspaces

Reviewer 1 workspace:  scVI Human Validation — Reviewer 1 (Blinded)

Reviewer 2 workspace:  scVI Human Validation — Reviewer 2 (Blinded, paginated)

Workflow

Assignment: share each reviewer document only with its assigned reviewer. Do not share this consensus document until both independent formal reviews are locked.

Calibration: both reviewers complete CAL-01 to CAL-03 independently, discuss them, and freeze the codebook before formal validation.

Independent review: reviewers complete all 50 HV records in their own different orders without viewing the other document.

Lock: record completion outside the document and prevent answer changes before reconciliation.

Reconciliation: copy both locked records into the matching HV row below; discuss disagreements only; enter the consensus result.

Unresolved reference: use `REFERENCE_UNRESOLVED` when the two reviewers cannot agree. Never force a label.

Analysis boundary: use the private local sampling crosswalk only after consensus is frozen. The 40 machine-classified studies support weighted accuracy metrics; the 10 abstention states are a separate process audit.

Locked record format

Citation validity: Yes / No / Unclear

Method executed: Yes / No / Unclear

Use labels (select all that apply): Benchmark / Extend or develop / Apply to biological data / Other

Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear

Biological insight: Yes / No / Unclear

Evidence sufficiency: Sufficient / Incomplete / Citation not located

Supporting evidence: exact quote plus section, page, paragraph, figure, or table locator.

Consensus status: AGREED / RECONCILED / REFERENCE_UNRESOLVED

Consensus table

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
CAL-01 CALIBRATION Cell Subtype Classification via Representation Learning Based on a Denoising Autoencoder for Single-Cell RNA Sequencing 2021 IEEE Access DOI: 10.1109/access.2021.3052923 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
CAL-02 CALIBRATION Eleven grand challenges in single-cell data science 2020 Genome biology DOI: 10.1186/s13059-020-1926-6 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
CAL-03 CALIBRATION Mosaic integration of spatial multi-omics with SpaMosaic 2024 DOI: 10.1101/2024.10.02.616189 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4 Supplement 5 Supplement 6	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-001 FORMAL VALIDATION Computational tools for analyzing single-cell data in pluripotent cell differentiation studies 2021 Cell Reports Methods DOI: 10.1016/j.crmeth.2021.100087 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-002 FORMAL VALIDATION Multimodal Hypersampling of the adult human heart resolves tissue-specific cell states and disease-associated variants 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.16.732617 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-003 FORMAL VALIDATION BrainBridge Characterizes Key Factors affecting Alzheimer's Disease and Associated Phenotypes 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2025.12.03.692255 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-004 FORMAL VALIDATION Hybrid deep learning–mechanistic modeling of cellular dynamics from a spatiotemporal single-cell atlas 2025 DOI: 10.1101/2025.10.02.679951 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-005 FORMAL VALIDATION Single-cell integration and multi-modal profiling reveals phenotypes and spatial organization of neutrophils in colorectal cancer. 2026 DOI: 10.1016/j.ccell.2025.12.003 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-006 FORMAL VALIDATION Implementing distinct spatial proteogenomic technologies: opportunities, challenges, and key considerations 2024 Clinical & Experimental Immunology DOI: 10.1093/cei/uxae077 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-007 FORMAL VALIDATION Biological network-inspired interpretable variational autoencoder 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.12.17.423310 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-008 FORMAL VALIDATION LAIOR: a hyperbolic neural ODE variational framework for interpretable single-cell manifold learning and trajectory inference 2026 Frontiers in Genetics DOI: 10.3389/fgene.2026.1838613 Article page Full text 1 Supplement 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-009 FORMAL VALIDATION Coding agents author interpretable single-cell embedding models from the literature 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.07.07.737048 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-010 FORMAL VALIDATION Single-cell characterization of anti-LAG-3 and anti-PD-1 combination treatment in patients with melanoma 2023 Journal of Clinical Investigation DOI: 10.1172/jci164809 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-011 FORMAL VALIDATION JIND-Multi: Leveraging Multiple Labeled Datasets for Automated Annotation of Single-Cell RNA and ATAC Data 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.01.15.633130 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-012 FORMAL VALIDATION sCellIST: a Multiple Instance Learning approach to predict single-cell gene expression from H&E images using spatial transcriptomics 2024 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.11.07.622225 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-013 FORMAL VALIDATION COVID-Datathon: Biomarker identification for COVID-19 severity based on BALF scRNA-seq data 2021 arXiv (Cornell University) DOI: 10.48550/arxiv.2110.04986 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-014 FORMAL VALIDATION Kinetic Foundation of the Zero-Inflated Negative Binomial Model for Single-Cell RNA Sequencing Data 2020 SIAM Journal on Applied Mathematics DOI: 10.1137/19m1253198 Article page Full text 1 Full text 2	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-015 FORMAL VALIDATION Recovering Single-cell Heterogeneity Through Information-based Dimensionality Reduction 2021 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2021.01.19.427303 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-016 FORMAL VALIDATION AtlasLens: Metadata-centric exploration and analysis of single-cell atlases 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.07.10.735823 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-017 FORMAL VALIDATION Multi-omic identification of perineurial hyperplasia and lipid-associated nerve macrophages in human polyneuropathies 2025 Nature Communications DOI: 10.1038/s41467-025-62964-8 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-018 FORMAL VALIDATION Hierarchical classification of immune cell transcriptomes at population-scale 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.05.30.728980 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-019 FORMAL VALIDATION Anti-bias training for (sc)RNA-seq: experimental and computational approaches to improve precision 2021 Briefings in Bioinformatics DOI: 10.1093/bib/bbab148 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-020 FORMAL VALIDATION SCIntRuler: guiding the integration of multiple single-cell RNA-seq datasets with a novel statistical metric 2024 Bioinformatics DOI: 10.1093/bioinformatics/btae537 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-021 FORMAL VALIDATION Biologically Informed Variational Inference Enables Interpretable Cell Phenotyping and Discovery 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.06.10.657924 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-022 FORMAL VALIDATION Transforming microfluidics for single-cell analysis with robotics and artificial intelligence 2025 Lab on a Chip DOI: 10.1039/d5lc00216h Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-023 FORMAL VALIDATION Transcriptomic advances in studies of muscle stem cell aging: From bulk to single-cell and beyond 2026 Cell Research DOI: 10.1038/s41422-026-01240-w Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-024 FORMAL VALIDATION A multi-dataset single-cell meta-analysis of human keloid skin across Asian, Black and White populations reveals endothelial and mesenchymal programs linked to ethnic disparities 2026 Frontiers in Medicine DOI: 10.3389/fmed.2026.1751725 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-025 FORMAL VALIDATION PI3Ky in B cells promotes antibody responses and generation of antibody-secreting cells 2024 Nature Immunology DOI: 10.1038/s41590-024-01890-1 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4 Supplement 5 Supplement 6 Supplement 7 Supplement 8 Supplement 9 Supplement 10	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-026 FORMAL VALIDATION Tumor-infiltrating CD27 - IgD - regulatory B cells suppress cytotoxic CD8 + T cell responses in renal cell carcinoma 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.07.08.663720 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-027 FORMAL VALIDATION Multi-modal Diffusion Model with Dual-Cross-Attention for Multi-Omics Data Generation and Translation 2025 DOI: 10.1101/2025.02.27.640020 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-028 FORMAL VALIDATION Deep learning applications in single-cell genomics and transcriptomics data analysis 2023 Biomedicine & Pharmacotherapy DOI: 10.1016/j.biopha.2023.115077 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-029 FORMAL VALIDATION Relevant Applications of Generative Adversarial Networks in Drug Design and Discovery: Molecular De Novo Design, Dimensionality Reduction, and De Novo Peptide and Protein Design 2020 Molecules DOI: 10.3390/molecules25143250 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-030 FORMAL VALIDATION Epigenetic biomarkers in inflammatory bowel diseases—computational challenges and opportunities 2026 Journal of Crohn s and Colitis DOI: 10.1093/ecco-jcc/jjag022 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-031 FORMAL VALIDATION Candida albicans exhibits distinct cytoprotective responses to anti-fungal drugs that facilitate the evolution of drug resistance 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.01.21.914549 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-032 FORMAL VALIDATION Effectively Clustering Single Cell RNA Sequencing Data by Sparse Representation 2021 IEEE Transactions on Computational Biology and Bioinformatics DOI: 10.1109/tcbb.2021.3128576 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-033 FORMAL VALIDATION HoloCell: A Generative Foundation Model for Holistic Cellular Modeling 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.07.730684 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-034 FORMAL VALIDATION Single-Cell Multi-Omics: Insights into Therapeutic Innovations to Advance Treatment in Cancer 2025 International Journal of Molecular Sciences DOI: 10.3390/ijms26062447 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-035 FORMAL VALIDATION Sequencing dropout-and-batch effect normalization for single-cell mRNA profiles: a survey and comparative analysis 2020 Briefings in Bioinformatics DOI: 10.1093/bib/bbaa248 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-036 FORMAL VALIDATION SECANT: a biology-guided semi-supervised method for clustering, classification, and annotation of single-cell multi-omics 2022 PNAS Nexus DOI: 10.1093/pnasnexus/pgac165 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-037 FORMAL VALIDATION Assessing Markovian and Delay Models for Single-Nucleus RNA Sequencing 2023 Bulletin of Mathematical Biology DOI: 10.1007/s11538-023-01213-9 Article page Full text 1 Supplement 1 Supplement 2	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-038 FORMAL VALIDATION Deep-learning methods for unveiling large-scale single-cell transcriptomes 2024 Cancer Biology and Medicine DOI: 10.20892/j.issn.2095-3941.2023.0436 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-039 FORMAL VALIDATION SCOPE: Localizing fate-decision states and their regulatory drivers in single-cell differentiation 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.04.07.717037 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-040 FORMAL VALIDATION Best practices for single-cell analysis across modalities 2023 Nature Reviews Genetics DOI: 10.1038/s41576-023-00586-w Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-041 FORMAL VALIDATION Decoding disease and therapy through multiomics integration and systems analysis 2026 Briefings in Bioinformatics DOI: 10.1093/bib/bbag311 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-042 FORMAL VALIDATION Pertpy: an end-to-end framework for perturbation analysis 2025 Nature Methods DOI: 10.1038/s41592-025-02909-7 Article page Full text 1 Supplement 1 Supplement 2 Supplement 3 Supplement 4	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-043 FORMAL VALIDATION Benchmarking atlas-level data integration in single-cell genomics 2021 Nature Methods DOI: 10.1038/s41592-021-01336-8 Article page Full text 1 Full text 2	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-044 FORMAL VALIDATION Toward universal cell embeddings: integrating single-cell RNA-seq datasets across species with SATURN 2024 Nature Methods DOI: 10.1038/s41592-024-02191-z Article page Full text 1 Full text 2	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-045 FORMAL VALIDATION An overview of computational methods in single-cell transcriptomic cell type annotation 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf207 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-046 FORMAL VALIDATION Projecting single-cell transcriptomics data onto a reference T cell atlas to interpret immune responses 2020 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2020.06.23.166546 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-047 FORMAL VALIDATION Deep learning in single-cell and spatial transcriptomics data analysis: advances and challenges from a data science perspective 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf136 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-048 FORMAL VALIDATION Beyond the List: Bioagent-Agnostic Signatures Could Enable a More Flexible and Resilient Biodefense Posture Than an Approach Based on Priority Agent Lists Alone 2021 Pathogens DOI: 10.3390/pathogens10111497 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-049 FORMAL VALIDATION Trends and Potential of Machine Learning and Deep Learning in Drug Study at Single-Cell Level 2023 Research DOI: 10.34133/research.0050 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes:
HV-050 FORMAL VALIDATION Batch correcting single-cell spatial transcriptomics count data with Crescendo improves visualization and detection of spatial gene patterns 2025 Genome biology DOI: 10.1186/s13059-025-03479-9 Article page Full text 1	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Use labels (select all that apply): Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Rationale/notes: